**Two-*(CL) structures for the price of one-(CL) in spoken Persian**

**Overview.** Numerals within nominal arguments are often analyzed as either dependents of a functional category (1a) or heads on the nominal spine (1b) (see e.g. Danon 2012). We propose that certain numerals in spoken Persian can occupy either structure, with left-branch extraction (LBE) available only in (1a)-type structures. We derive this contrast and additional facts in terms of featural selection.

(1) a. $\left[_{\text{FP}}\ \text{NumP}\ [_{\text{NP}} \ldots ]\right]$ b. $\left[_{\text{NumP}}\ \text{Num}^0\ [_{\text{NP}} \ldots ]\right]$

**Numeral classifier puzzle.** Most numerals in spoken Persian must be followed by (i) a true classifier (abbreviated TCL, viz. *tā*), or (ii) a (semi-lexical) sortal or measure classifier (abbreviated MCL, e.g. *šāxe* 'branch,' *baste* 'pack,' *ja'be* 'box') and allow both kinds of classifiers to be present, provided that the TCL precedes the MCL (2b). However, the numeral *yek* 'one' is incompatible with TCLs, and allows — but does not require — an MCL (2a). Nouns in both cases are marked singular (outside of exceptional constructions where the plural marker takes on other functions, cf. Ghomeshi 2003; Shirzad & Darzi 2023:154).

(2) a. *ye(k) (*tā) (šāxe) gol*
one TCL MCL flower
'one flower'

b. *do (tā) šāxe gol(*-hā)*
*do* *(*tā*) *gol*(*-*hā*)
two TCL MCL flower(-PL)
'two flowers'

The incompatibility of TCLs and *yek* is not simply due to the singular semantics of the numeral 'one', and the optionality of MCLs with *yek* is not because it can be parsed as an indefinite article. Both explanations can be ruled out by an underreported fact about this paradigm: the distribution of *yek* shown above is shared with all conjoined numerals with *yek* in the ones place, e.g. *si-o-yek* (lit. 'thirty-and-one'). Regardless of the internal structure of such conjunctions, the external distribution shows that these numerals differ formally from others; we thus distinguish *Type-I numeral*s (with the *yek* distribution, 2a) from *Type-II numerals* (2b).

Prior work has suggested that numerals, MCLs, and TCLs are all heads on the nominal spine (Gebhardt 2009; Shirzad & Darzi 2023, a.o.). This would suggest that these *numeral + classifier complexes* (NCCs) are not constituents, but rather form constituents with the modified nouns. However, this is contradicted by widely unnoticed facts, e.g. that NCCs can serve as fragment answers and, more strikingly, that spoken Persian permits left-branch extraction (LBE) of NCCs, where the rest of the nominal remains *in situ* (3). Note that the numeral here may be information-structurally marked, suggesting pied-piping of the classifier.

(3) [*ye(k) ja'be*]$_i$ *man har ruz* [___$_i$ *širini*] *mi-xor-am*
one MCL 1SG every day sweet DUR-eat:PRS-1SG
'I eat *one* box of sweets every day.'

We take this to be *bona fide* movement, because this discontinuity is sensitive to CED islands. For example, (4), where *qazā* is the complement of the verb, is degraded because the gap is within a specifier.

(4)$^{??}$[*do tā*]$_i$ *man ruzi* [___$_i$ *gorbe*] *qazā mi-d-am*
two TCL 1SG per_day cat food DUR-give:PRS-1SG
Intended: 'I give *two* cats food per day.'

So NCCs are clearly moveable constituents, but what about the bare Type-I numerals (2a)? Whereas these numerals do not need a classifier, without one, they curiously cannot undergo LBE, as demonstrated in (5).

(5) **ye(k)*$_i$ *man har ruz* [___$_i$ *širini*] *mi-xor-am*
one 1SG every day sweet DUR-eat:PRS-1SG
Intended: 'I eat *one* sweet every day.'

Two issues thus require explanation w.r.t. spoken Persian numerals: (i) the distribution of numerals/classifiers within nominals and (ii) the variability in what can undergo LBE. In this case study, we show how all the relevant facts can be explained in terms of the interaction between selection and projection.

**Assumptions.** We assume that Merge is licensed by checking matching features under sisterhood (Chomsky 1995; Adger 2003; Heck & Müller 2007; Zeijlstra 2020, a.m.o.). Critically, following Zyman (submitted) and *Author* (2025), we assume that selectional features are separated into those that trigger projection of the selecting element (standard c-selection, 6a) and those that do not (e.g. selection of hosts by adjuncts, 6b).

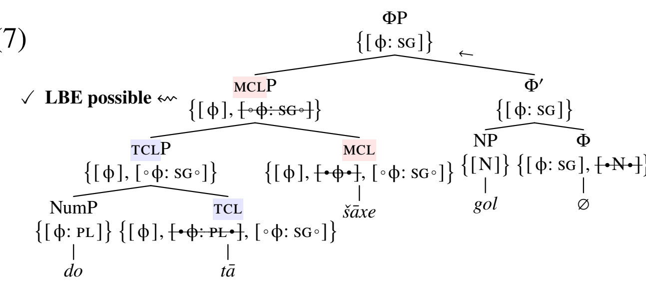
(6) a. If $\alpha$ bears $[\bullet f \bullet]$ and $\beta$ bears $[f]$, then $\{\alpha, \beta\}$ is licensed s.t. it inherits only $\alpha$'s other features.

b. If $\alpha$ bears $[\circ f \circ]$ and $\beta$ bears $[f]$, then $\{\alpha, \beta\}$ is licensed s.t. it inherits only $\beta$'s features.

This gives us probes for Merge, notated $[\bullet f \bullet]$ and $[\circ f \circ]$ — which, when discharged, result in different projection profiles — and goals, notated $[f]$. Following Preminger (2014) and Newman (2021), a.o., we assume that probes for Merge/Agree must be used where possible, but may unproblematically fail elsewhere. Finally, following Zeijlstra (2020), we assume that probes and goals for Merge may optionally be specified for subfeatures/values, similar to those for Agree (cf. Pesetsky & Torrego 2007).

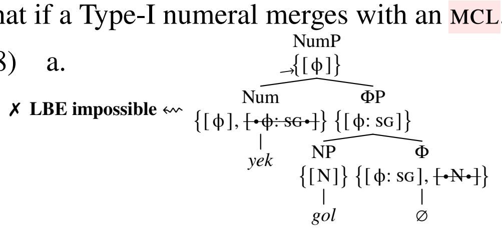
**Type-II analysis.** With these assumptions, we analyze NCCs as adjoining to a functional $\Phi$-shell (cf. Bale et al. 2019) whose head bears the feature $[\phi\text{: SG}]$, in complementary distribution with the head spelled out as the plural marker (bearing $[\phi\text{: PL}]$, person values are orthogonal here). Type-II numerals themselves bear only $[\phi\text{: PL}]$, and $\Phi$P bears no probes, so classifiers are used to connect the two. TCLs bear an unvalued $[\phi]$ and can select Type-II numerals as complements using a probe $[\bullet\phi\text{: PL}\bullet]$. MCLs also bear $[\phi]$, but bear the unspecified probe $[\bullet\phi\bullet]$. The lack of a value crucially allows MCLs to merge either with a Type-II numeral (with $[\phi\text{: PL}]$) *or* with a TCL (with $[\phi]$). Both MCLs and TCLs also bear the probe $[\circ\phi\text{: SG}\circ]$, which allows them to adjoin to the $\Phi$-shell. As $[\circ\phi\text{: SG}\circ]$ prevents classifiers from projecting (6b), NCCs become subconstituents of $\Phi$P, and can undergo LBE (3). We stipulate that classifiers are linearized rightward (like the DOM marker *rā* and the plural *-hā*); as MCLs can select TCLs as complements but not *vice versa*, this then correctly predicts that TCLs must precede MCLs if they co-occur; this whole derivation is sketched in (7).

(7)

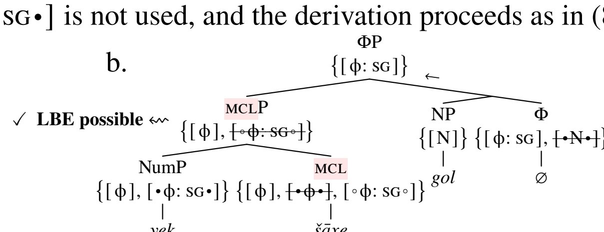



**Type-I analysis.** Type-I numerals show a different distribution: they are compatible with MCLs but do not require them, and are altogether incompatible with TCLs. On this approach, they differ featurally from Type-II numerals in two ways. First, they bear the underspecified feature $[\phi]$, making them suitable complements for MCLs with $[\bullet\phi\bullet]$ (8b), but not for TCLs, which have the more specified probe $[\bullet\phi\text{: PL}\bullet]$. In other words, the unvalued $\phi$ — which gives MCLs the flexibility to select either Type-II numerals or TCLs — also renders Type-I numerals incompatible with TCLs. Second, we reason that they also bear their own probe $[\bullet\phi\text{: SG}\bullet]$, which generates an alternate structure where the numeral selects $\Phi$P as its complement and projects its own $[\phi]$ feature (8a). This solves two problems at once: first, a classifier is no longer necessary, and second, the numeral now forms a constituent with $\Phi$P, correctly ruling out LBE (see 5). This structure can then be targeted by any Merge/Agree probe for $\phi$-features (both $[\phi\text{: SG}]$ and $[\phi]$ control 3SG/default verb agreement, *modulo* semantic agreement). Note that if a Type-I numeral merges with an MCL, $[\bullet\phi\text{: SG}\bullet]$ is not used, and the derivation proceeds as in (8b).

(8) a.




b.




**Discussion.** In summary, the featural analysis above accounts for five key facts: (1) either a TCL or an MCL must be present for Type-II numerals to be licensed; (2) if both are present, then the TCL must precede the MCL; (3) NCCs are constituents which can undergo LBE; (4) Type-I numerals do not require a classifier, but may be followed only by an MCL; (5) In the absence of an MCL, LBE of a Type-I numeral is impossible.

Zooming out, the approach taken here is to model overlapping distributions using feature values and projectional differences in lieu of arbitrary probes/goals for every category (e.g. [TCL], [MCL], etc.). A good result of this is that $[\phi]$ may project from the numeral rather than $\Phi$P, blocking LBE without affecting the distribution of the nominal. More broadly, since LBE itself is essentially counterfed by projection, this also raises the prospect that typologically, projectional variation of the kind anticipated by this analysis may contribute to variability in which adnominal elements can undergo LBE within a given language.

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